ALGEBRA 2 · UNIT 5: RADICALS & RATIONAL EXPONENTS

Crash Scene Investigator

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students act as accident reconstruction analysts. They use the skid-mark formula s = √(30df) to estimate speeds, simplify radicals, solve radical equations and check for extraneous solutions, graph square-root functions, and use rational exponents with pendulums and planetary orbits.

Radicals model real science formulas. Solving for skid length or orbit size makes students square both sides or use rational exponents, and the extraneous-solution check becomes part of the evidence.

Standards Addressed

- N-RN.1–2 Rational exponents and radicals
- A-REI.2 Solve radical equations and identify extraneous solutions
- F-IF.7b Graph square root functions
- A-CED.4 Rearrange formulas

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with the case file. Part 1 skid formula and simplifying.
Day 2	Part 2 solving and graphing.
Day 3	Part 3 pendulums, orbits, and the report.

Materials

- Student handout
- Scientific calculators
- Measuring tape (optional skid-length activity)

How the Three Versions Differ

Level	What changes
CP2	Evaluate and simplify radicals, solve one radical equation with a check, and use a pendulum formula.
CP1	Rearrange formulas, graph square-root transformations, and use rational exponents with Kepler's law.
Honors	Sensitivity analysis, equations with two radicals, and solving for orbit size with ⅔ powers.

Watch For (Common Misconceptions)

- Forgetting to check for extraneous solutions.
- Writing $\sqrt{(a + b)} = \sqrt{a} + \sqrt{b}$.
- Confusing the ½ power with dividing by 2.

Extension Ideas

- Measure stopping distances with a bike (safely).
- Research how friction factors are measured.
- Compare planetary data from NASA.

Answer Key – CP2

Part 1: The Skid Marks

#	Task	Answer
1	Find the speed for d = 120 ft on dry asphalt (f = 0.75). Write the exact answer in simplest radical form and as a decimal.	$\sqrt{(30 \cdot 120 \cdot 0.75)} = \sqrt{2700} = 30\sqrt{3} \approx 51.96$ mph
2	Was the driver speeding in the 45-mph zone?	Yes, about 52 mph in a 45-mph zone.
3	Find the speed for the same skid on wet asphalt (f = 0.45). Simplify the radical.	$\sqrt{(30 \cdot 120 \cdot 0.45)} = \sqrt{1620} = 18\sqrt{5} \approx 40.25$ mph
4	How long would the skid be on dry asphalt at exactly 45 mph? Solve $45 = \sqrt{(22.5d)}$.	$2025 = 22.5d \rightarrow d = 90$ ft

Part 2: Solve & Check

#	Task	Answer
5	Solve $\sqrt{(x + 7)} = x - 5$. Check both answers and identify any extraneous solution.	$x + 7 = x^2 - 10x + 25 \rightarrow x^2 - 11x + 18 = 0 \rightarrow x = 9$ or 2. Check: x = 9: $\sqrt{16} = 4 = 4$ ✓. x = 2: $\sqrt{9} = 3$ but $2 - 5 = -3$ ✗, so 2 is extraneous.
6	Explain why squaring both sides can create an answer that doesn't work.	Squaring makes −3 and 3 both become 9, so a false equation can become true.
7	Make a table for $s(d) = \sqrt{(22.5d)}$ with d = 0, 40, 90, 160 and sketch the graph.	0, 30, 45, 60; the graph rises quickly and then levels off.

Part 3: Pendulums & Report

#	Task	Answer
8	Find the period of an 8-ft pendulum.	$2\pi\sqrt{(8 \div 32)} = 2\pi(0.5) = \pi \approx 3.14$ s
9	Find the period of a 2-ft pendulum. How does making the pendulum 4 times shorter change the period?	$2\pi\sqrt{(2 \div 32)} = 2\pi(0.25) = \frac{1}{2}\pi \approx 1.57$ s; the period is cut in half.
10	Write a short report on the crash case using your answers.	Answers vary.

Answer Key – CP1

Part 1: The Skid Marks

#	Task	Answer
1	Find the dry-road speed for a 120-ft skid in simplest radical form. Then write the formula using a rational exponent.	$30\sqrt{3} \approx 51.96$ mph; $s = (30df)^{1/2}$
2	Solve the formula for d. Use it to find the skid length at 45 mph and at 60 mph on dry asphalt.	$d = s^2 \div (30f)$. 45 mph: $2025 \div 22.5 = 90$ ft. 60 mph: $3600 \div 22.5 = 160$ ft.
3	A second skid is 4 times as long on the same road. Show algebraically that the speed is 2 times as large.	$\sqrt{(30 \cdot 4d \cdot f)} = \sqrt{4} \cdot \sqrt{(30df)} = 2s$

Part 2: Solve & Graph

#	Task	Answer
4	Solve $\sqrt{(x + 7)} = x - 5$ and identify any extraneous solution.	$x = 9$ ($x = 2$ is extraneous)
5	Solve $\sqrt[3]{(2x - 3)} = 3$.	$2x - 3 = 27 \rightarrow x = 15$
6	Describe $s(d) = \sqrt{(22.5d)}$ as a transformation of $y = \sqrt{d}$. Graph both.	Vertical stretch by $\sqrt{22.5} \approx 4.74$ (or horizontal compression by 22.5).
7	On a wet road, $s(d) = \sqrt{(13.5d)}$. Is this graph above or below the dry graph? Why?	Below: $13.5d < 22.5d$ for $d > 0$, so less friction means lower speed estimates for the same skid.

Part 3: Orbits & Report

#	Task	Answer
8	Mars has a = 1.52 AU. Find its orbit time.	$1.52^{1.5} \approx 1.87$ years
9	Saturn's orbit takes 29.5 years. Solve for a (use the ⅔ power) and find a.	a = 29.5 to the ⅔ power ≈ 9.55 AU
10	Write the case report and include one radical and one rational-exponent result.	Answers vary.

Answer Key – Honors

Part 1: Evidence Analysis

#	Task	Answer
1	If the skid length is measured 10% too short, by what percent is the speed underestimated? Use exponent properties.	True d is d ÷ 0.9, so the true speed is s ÷ √0.9 ≈ 1.054s: about 5.1% higher than estimated (≈ 4.9% underestimate).
2	The driver claims 45 mph. On a road with an unknown friction factor, what f would make the 120-ft skid match the claim?	$2025 = 30 \cdot 120 \cdot f \rightarrow f = 0.5625$
3	Write s as a power function of d and describe how the speed grows.	s = (22.5d) raised to the ½ power = √22.5 · d raised to the ½ power: speed grows like the square root of the skid length.

Part 2: Harder Equations

#	Task	Answer
4	Solve √(x + 6) − √(x − 1) = 1.	√(x + 6) = 1 + √(x − 1) → x + 6 = x + 2√(x − 1) → 3 = √(x − 1) → x = 10. Check: 4 − 3 = 1 ✓
5	Solve x to the ⅔ power = 9 (that is, the cube root of x^2, squared form) for real x.	$x^2 = 729 \rightarrow x = \pm 27$ (both check because x^2 is positive)
6	Solve √(2x + 1) = x − 1 and identify extraneous solutions.	$2x + 1 = x^2 - 2x + 1 \rightarrow x^2 - 4x = 0 \rightarrow x = 0$ or 4. x = 0: √1 = 1 but 0 − 1 = −1 ×. Only x = 4.

Part 3: Orbits & Report

#	Task	Answer
7	Jupiter's orbit takes 11.86 years. Find a.	a = 11.86 to the ⅔ power ≈ 5.20 AU
8	Show that doubling a multiplies T by about 2.83. Write the exact factor.	$(2a)^{1.5} = 2^{1.5} \cdot a^{1.5} = 2\sqrt{2} \cdot T \approx 2.83T$
9	Write the full report, including the error analysis.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Conclusions are supported by correct radical calculations, and any extraneous solutions are identified. Goes beyond the minimum with a creative, realistic choice.	Conclusions are supported by correct radical calculations, and any extraneous solutions are identified.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes speed estimates with simplified radicals, solved formulas, a graph, and a conclusion about the case. Polished, organized, and visually engaging.	Report includes speed estimates with simplified radicals, solved formulas, a graph, and a conclusion about the case. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.